

Energy policy for low carbon development in Nigeria: A LEAP model application



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ABSTRACT

This paper applied a scenario-based analysis to explore Nigeria's future energy demand, supply and associated GHG emissions from 2010 to 2040 using the Long-range Energy Alternative Planning (LEAP) model. The impact of different energy policies are analysed for the Nigerian energy system by considering four scenarios: the reference scenario (REF), the low-carbon moderate scenario (LCM), the low-carbon advanced scenario (LCA), and the green optimistic scenario (GO). By considering aggressive energy policies and strategies from LCM to LCA, and even more aggressive options in the GO scenario, we find that under the REF scenario energy demand is expected to reach 3,075 PJ and a corresponding increase in GHG emissions of 201.2 Mt Co2e by 2040. More aggressive policy intervention by the Nigerian government, as in the GO scenario, would lead to a decrease in energy demand (2,249 PJ) and GHG emissions (124.4 Mt Co2e) in 2040. A cost-benefit and energy system analysis were also carried out in the study.

1. Introduction

During the pre-industrial revolution, the global development was hindered by low energy intensity in the form of hydro and wind resources. The second industrial revolution saw the increase in fossil fuel consumption which lead to electrification and technological commercialization of goods and services [48]. From last century, the world has metamorphosed and experienced a phenomenal transition in the way energy is used, from coal-based to petroleum-based. Ever-increasing globalisation and industrialisation have exponentially increased the demand for energy worldwide [43]. To cope with this exponential increase in demand for energy, energy production has proportionately increased to such an extent that approximately 80% of global energy supply comes from fossil fuel (IEA, 2008). The increase in dependence on fossil fuels has resulted in an increase in global greenhouse gas emissions, which has raised issues about the sustainability of our environment due to tribulations such as climate change and the depletion of natural resources [47].

Energy consumption has been one of the most unswerving indicators of development and quality of life attained by any country, and the necessity of satisfying a forecasted energy demand for a given period of

time is the key rationale for energy planning [7]. As defined by the World Energy Council, “Energy planning is that part of economics applied to energy problems, taking into account the analysis of energy supply and demand, as well as implementation of the means for ensuring coverage of energy needs in a national or international context” [11]. Researchers around the world have employed different energy models to address policy and planning concerns of energy, economy, and the environment [35].

Access to clean energy is unreliable and has high disruption costs, affecting production efficiency and competitiveness in many developing countries in Africa and Southeast Asia [18]. Despite being gifted with the widest possible range of energy resources, the African continent has experienced a relatively low energy consumption in general, and electricity consumption in particular (Mayo, 2012). Nigeria is experiencing a remarkable paradox—the abundance of energy resources and widespread energy poverty. About 40% of the population have access to the grid electricity supply, while 70% depend on firewood even to this day [13]. This dependence on firewood constitutes a major indoor pollution hazard and has resulted in the death of nearly 79,000 Nigerians due to smoke inhalation in 2011 [12]. According to a study by the World Health Organization in 2013, the deaths caused by smoke

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inhalation from firewood used by women has reached 98,000 [15].

The Nigerian government's response to the issue of energy poverty has been to increase the number of gas power plants for electricity generation [16], while plans are still in progress to introduce other sources of energy. The plans are being set up by the Energy Commission of Nigeria (ECN), a government agency which is responsible for the strategic planning and coordination of national energy policies in all its various forms in Nigeria [19]. The ECN carried out a study to ascertain the projected future energy demand and supply in some scenarios. This study was carried out using the Model for the Energy Supply Strategy Alternatives and their General Environmental Impact (MESSAGE), and the results are included in the current National Energy Master Plan [17,30].

However, the study carried out by the ECN missed several important elements: (a) The ECN did not develop scenarios considering various parameters which can influence energy demand. The only parameter considered was economic growth rate (GDP), which is not the only determinant for the increase in energy demand. Other parameters include population, income, household size, technology, energy prices, etc. Also, minimum cost options were not observed in the ECN study. (b) The impact of future energy policy implementation and strategies were not considered in the ECN study. The impact of various energy policies is expected to alter the energy consumption pattern of a country on the short-, medium-, and long-term horizons. Also, incentives such as Feed-in-Tariff may affect energy consumption patterns. (c) The effects of greenhouse gases on the environment were not considered in the ECN study. The contribution of GHGs to climate change cannot be left out when carrying out energy modelling because this has a direct impact on the society. The effect of climate change can be observed in almost all parts of Nigeria, and as such the consideration of GHG reduction is important.

Our study is an attempt to address the following pertinent questions which would steer forward energy policy reforms in Nigeria in years to come: (a) What is the projected energy demand in Nigeria and how can this demand be met efficiently at minimum cost without increasing GHGs? (b) Which sustainable energy policy option can be recommended to ensure that low-carbon development is realised in Nigeria? The study aims to answer the research questions by using the Long-Range Energy Alternatives Planning (LEAP) system.

It is anticipated that this investigation will present an opportunity for Nigeria to select from various renewable energy options, low-carbon technologies, and energy efficiency tools to moderate growth in energy demand. This study also intends to provide vital energy policy recommendations for the Nigerian government and energy policy experts to strengthen planning for the future energy systems in Nigeria. This study, however, does consider the ability of consumers to pay for energy services or explore the effect of energy pricing. The scope of this study lies within the range of projected energy demand, supply, and its accompanying GHGs reductions, while briefly projecting the social cost of each policy scenario in the Nigerian LEAP model.

The rest of the paper is organised as follows. Section 2 describes the methodology used in this study elaborating the model, formulation of policy scenarios, and the relevant data used with the source. The results and discussion of the scenarios developed are presented in Section 3, while Section 4 concludes the paper with recommended policy implications based on our study.

2. Methodology, data and scenario development

2.1. The model

The Long-Range Energy Alternatives Planning (LEAP) system [27] is an integrated modelling tool used for energy policy analysis and climate change mitigation assessment [41]. It was developed by the Stockholm Environment Institute, and can be used to develop various scenarios of projected energy demand and environmental impact based

on how energy is consumed, transformed, and generated in a given region or economy under a range of values for parameters such as population increase, gross domestic product, income, etc. [5]. The LEAP model has a flexible data structure which is not only easy to use, but also rich in technical and end-user details [45]. It has been extensively adopted in many organisations on the local, national, and international levels to project energy supply and demand, predict environmental impact of energy policies, and identify possible challenges in the future.

In the household, commercial, transport and industrial sectors around the world, the LEAP model have been used to assess their energy consumption and greenhouse gas emissions [37]. Studies such as Bose [4] simulated the passenger transport sector for New Delhi city in India, while Shin et al. [39] analysed the impact of expanded land fill gas (LFG) method of electricity generation on cost of electricity generated and GHG emission in South Korea. The environmental and economic assessment of energy policy for South Korea was carried out by Song et al. [40] using the LEAP model. The energy sector in Estonia was modelled for environmental impact of electricity generation from newer, lesser amount of environmental destructive technology using the LEAP model [9]. Furthermore, Huang et al. [22] has elucidated the importance of long-term forecasting of energy supply and demand for Taiwan. Scenarios were developed to present a possible pathway to a low-carbon economy for China in Taoa et al. [44] study, and this low-carbon pathway was further developed in Feng and Zhang [20] study for Beijing. The LEAP has been applied to ascertain the projected direct energy consumption in Nigerian households [23], while it has been used for other African countries such as Ethiopia [38] who developed alternative scenarios for the country's energy requirement up-to 2050. The energy demand and pollutant emissions for the Greek road transport sector was analysed by Bitos and Kiartzis [3] from 2010 to 2035. More recently, Mahumane and Mulder [28] estimated the aggregate trends in energy supply and demand due to projected gush in natural resource exploration in Mozambique. Other studies that have applied the LEAP model include; Kadian et al. [24] for the household sector, Dhakal [10] and Pradhan et al. [36] for the transport sector in India and Nepal respectively.

Several studies in literature have focused on mitigation of GHG emission by shifting the energy utilization in demand side, while maximizing the potentials of diversification of the energy supply mix. This study falls into category where it intends to lessen GHG emissions by imposing policies in both supply and demand side for Nigeria energy system in a bottom-up approach manner through the use of the LEAP model. However, the only bottom-up study that comprehensively investigated the impact of energy policies in the Nigerian household was carried out by Ibitoye [23]. Thus, no other study has applied a bottom-up approach to ascertain the impact of energy policies in the future energy system in Nigeria and this study intends to fill the gap in the literature.

2.2. The algorithm of the LEAP model

The LEAP model uses a framework for calculating energy consumption, transformation (electricity generation, oil refinery, charcoal production, coal mining), and carbon emissions. The values are imputed into the LEAP model based on the required data which may be available to the user. These are presented in the following sections.

2.2.1. Energy consumption

The total final energy consumption is calculated as follows [20]:

$$EC_n = \sum_i \sum_j AL_{n,j,i} \times EI_{n,j,i} \quad (1)$$

where EC is the aggregate energy consumption of a given sector, AL is the activity level and is a measure of the social or economic activity for which energy is consumed. AL is measured for each year as scale of

units per households or tonnes for industry. For example, to measure the activity level of households, the expression will be in the growth (%), the scale will be in millions or billions (depends on the population), and the units will be per households. The industry sector is the same in expression and scale, but measure in per metric tonnes (depending on the industrial output). In the transport sector, the units can be per person, passenger-km or tonne-km per person. EI is the energy intensity is the annual final consumption of energy per unit of activity level. In EI , the energy used in each sector (e.g. coal or gas) is measured in the units (e.g. megajoule or barrel) per metric tonne each year in the period under observation. n is the fuel type, i is the sector, and j is the device.

The net energy consumption for transformation is calculated as follows:

$$ET_s = \sum_m \sum_t ETP_{t,m} \times \left(\frac{1}{f_{t,m,s}} - 1 \right) \quad (2)$$

where ET is the net energy consumption for transformation,¹ ETP is the energy transformation product and is can be electricity in the case of power plants or refinery products (e.g. petrol, diesel, liquefied petroleum gas) in the case of crude oil refinery. f is the energy transformation efficiency and this is the energy content of the output fuels divided by the energy content of the feedstock fuels (measured in %). s is the type of primary energy, m is the equipment, and t is the type of secondary energy.

The transport stock turnover is calculated as follows:

$$EV_n = \sum_c s \times \frac{m}{fe} \quad (3)$$

where EV is the energy consumption in the transport sector, s is the number of vehicles (stock), m is the vehicle distance, fe is the fuel economy, n is the fuel type, and c is the vehicle type.

2.2.2. Transformation

The transmission and distribution module calculations take the domestic fuel requirement faced by the module and map those corresponding to the output fuels directly to the module input fuels. The total domestic requirements are then decreased by the outputs of the module and increased by the inputs to the module [26]:

For each process p :

$$INPUT_p = \frac{OUTPUT_p}{EFFICIENCY_p} \quad (4)$$

For a Transmission and Distribution module:

$$EFFICIENCY_p = 1 - LOSSES_p \quad (5)$$

Where $INPUT$ is the fuel or feedstock, $OUTPUT$ is the electricity generated or the refinery/production output, $EFFICIENCY$ is the efficiency of the power plants or refinery plants.

2.2.3. Carbon emission

The carbon emission from final energy consumption is calculated as follows [20]:

$$CEC = \sum_i \sum_j \sum_n AL_{n,j,i} \times EI_{n,j,i} \times EF_{n,j,i} \quad (6)$$

¹ The transformation sector comprises of the conversion of the primary energy to the secondary energy and further transformation. The transformation sectors include electricity power plants, petroleum refineries, coal ovens and blast furnaces. Further transformation includes electricity transmission and distribution centers, which delivers power to the final consumers (demand-side). The non-renewable energy (e.g. natural gas, coal, nuclear, etc.) from the supply-side are for natural gas turbine power plants, and coal for coal thermal power plants. Renewable energy such as wind energy for wind power plants, solar energy for solar photovoltaic, hydro resources for hydroelectric power plants, etc.

Where CEC is the carbon emission, AL is the activity level, EI is the energy intensity, and is the carbon emission factor from fuel type n for equipment j from sector i .

The carbon emission from energy transformation is calculated as follows:

$$CET = \sum_s \sum_m \sum_t ETP_{t,m} \times \frac{1}{f_{t,m,s}} \times EF_{t,m,s} \quad (7)$$

where CET is the carbon emission, ETP is the energy transformation product, f is the energy transformation efficiency, and is the emission factor from one unit of primary fuel type s consumed for producing secondary fuel type t through equipment m .

2.2.4. Costs

The total cost of sector is calculated [49]:

$$C = \sum_i \sum_j \left\{ \left[\sum_n (e_{n,j,k} ep_n) + \sum_k (mk_{j,i} mp_k) + fc_{j,i} \right] p_{j,i} \right\} \quad (8)$$

where C is the total cost of sector including equipment fixed costs and variable costs for raw materials and fuels, ep_n is the unit price of fuel type n , $mk_{j,i}$ is the demand for raw material k per unit of production used in equipment j within production process i , mp_k is the unit price of raw material k , and $fc_{j,i}$ is the fixed cost per unit production through equipment j (within production process i).

2.2.5. Cost-benefit analysis

The cost-benefit analysis² in LEAP calculates the costs of each part of the energy system, such as the capital and operating maintenance costs of purchasing and using the technologies in the Demand and Transformation systems; the costs of extracting primary resources and importing fuels; and the benefits from exporting fuels. Additionally, one can optionally broaden the scope of the cost-benefit calculations to examine environmental externalities by assigning costs to the emission of pollutants and any other direct social and environmental impacts of the energy system. LEAP performs cost-benefit calculations from a societal perspective by counting the costs in the energy system and then comparing the costs of any two scenarios [49].

2.3. Scenario development and assumptions

The scenarios to be formulated are based on the Nigerian government's planned expansion of electricity capacity, energy services, and proposed mitigation options for the reference (REF) scenario. Much of the formulation in the REF scenario is available in the National Energy Master Plan [19]. This research used the REF scenario as a benchmark for the development of three alternative scenarios, which are low-carbon moderate (LCM), low-carbon advance (LCA), and green optimistic (GO). The REF scenario represents the base case without any energy policy intervention. The alternative scenarios, on the other hand, have had their policy strategies increased in degrees ranging from the LCM with moderate policies, to a more aggressive LCA scenario, and finally the GO scenario, which has a high level of renewable energy application on both the demand and supply side. The summary of the key assumption parameters and features of each scenario used in the Nigerian LEAP model are presented in Table 1 and Table 2.

The Nigerian LEAP model is divided into two main branches (see

² A cost-benefit analysis is an analytical tool/technique used in determination of which options provide best possible approach in adoption/practice in terms of benefits in labour, time, and cost savings [8]. It's a systematic process observed in calculating and comparing the benefits and costs of a project, decision, or government policy which is aimed ascertaining if an investment decision is feasible, and to establish a basis for projects comparison ([6]; Policy, 2014) which includes comparing the total expected cost of each option against the total expected benefits [42].

Table 1

Key assumption parameters for the Nigerian LEAP model under the REF, LCM, LCA, and GO scenarios.

Key assumption	REF ^b		LCMS		LCAS		GOS	
	2010	2040	2010	2040	2010	2040	2010	2040
GDP (million USD)	369	1255 ^a	369	1476 ^a	369	1587 ^a	369	2177 ^a
GDP growth rate (%)	8		10		11		13	
Income (Thousand USD)	2310	5705	2310	5914	2310	6329	2310	6676
Income growth rate (%)	4.9		5.2		5.8		6.3	
Population (Million)	160	304	160	318	160	328	160	337
Population growth rate (%)	2.55	3	2.55	3.30	2.55	2.40	2.55	3.70
Households (Million)	37	72	37	80	37	84	37	87
Household size (people)	5							
Household growth rate (%)	3.16		3.9		4.2		4.5	
Urbanization rate (%)	44	70	44	75	44	80	44	73
Industry growth rate (%)	6	6.5	6	7.1	6	7.5	6	7.7
Commercial/ Services growth rate (%)	12	12.5	12	13.2	12	13.6	12	13.9
Agriculture growth rate (%)	6	6.5	6	7.3	6	7.4	6	7.8

^a denotes Billion.^b the figures in the REF scenario are based on the [30].

Fig. 1): the demand branch and the supply branch. The demand branch is made up of households, industry, commercial services, agriculture, and the transport sector. The household is further divided into urban and rural households which represent the current household situation in Nigeria. The households include various household appliances that are common in Nigerian households. The rural and urban households were further divided into electrified and non-electrified as stipulated by the National Bureau of Statistics [29]. The transport sector was also classified into six vehicle categories which include motorcycles, cars, light goods vehicles, heavy goods vehicles, urban buses, and long distance coaches. On the supply branch, the transformation sector was divided into transmission and distribution (for electricity and natural gas), electricity generation, charcoal production, oil refining, and coal mining. The resources branch under the transformation sector includes primary and secondary resources..

2.4. Data source and collection process

The Nigerian LEAP model was developed to forecast the future energy demand, supply, and GHG emission under the Reference scenario and alternative scenarios (explained in later subsections). The data collected for the Nigerian LEAP model were from many sources which includes the National Bureau of Statistics in Nigeria (NBS) (primary data source), Energy Commission of Nigeria (ECN) (for national policies, some primary data and plans), National Control Center, Osogbo in Nigeria (NCC) (power plants data), the Petroleum Products Pricing Regulatory Agency in Nigeria (PPPRA), the World Bank, International Energy Agency (IEA), the United States Energy Information Administration (EIA), the International Association of Public Transport (UITP) and African Association of Public Transport (UATP), National Surveys by bodies and independent researchers, credible research articles, newspaper and website articles and publications. The source of dataset used in this research are presented in Appendix 1.

The Nigerian LEAP dataset were vast but carefully divided into five subsectors of energy demand of which includes; households, industry, transport, commercial/service and agriculture. The household sectors was split into two main branches and two sub-branches under the main branches: the urban household was divided into electrified and non-electrified, while the rural household were divided into rural electrified and non-electrified. This is shown in Fig. 1 which is the Nigerian LEAP model data branches.

In the industry branch, energy use is divided into shares of energy use (i.e. gas, coal, electricity, biomass), and this is due to the data availability. These is also the situation in the commercial/service and

agriculture sectors. The transport branch is divided into six branches: motorcycles, cars, light goods vehicles, heavy goods vehicles, urban bus and long distance coach. The transformation branch is divided into five branches: transmission and distribution, electricity generation, charcoal production, oil refinery and coal mining. The final resource branch is the branch which includes the primary and secondary energy resources.

3. Results and discussion³

3.1. Results

3.1.1. Energy demand

Based on the assumptions about the socio-economic development in Nigeria and various parameters presented in Table 1 and Table 2, the values of total energy consumption the Nigerian LEAP model for the REF, LCM, LCA, and GO scenarios from 2010 to 2040 are shown in Fig. 2. In general, energy consumption will increase steadily up to 2040 under each scenario, but with different growth rates. The energy consumption in the REF scenario is expected to reach 3075 Petajoules (henceforth PJ) by 2040, from 1039 PJ in 2010, with an annual growth rate of 3.68%, the highest amongst the four scenarios. Much of the energy consumed in the REF scenario in the base year comes from the household sector with a share of 40%, which is followed by the industry (29%), and transport (22%). This, however, made a slight shift in 2040 with an increase in the household and commercial/service sector, while reduction was observed in the industry and transport sector..

Due to the series of energy efficiency improvements and fuel and technology switching, the total energy consumption will be lower under the alternative scenarios. The LCM scenario energy consumption is reduced to 2,941 PJ with an annual growth rate of 3.53%. As can be recalled from Table 2, the moderate improvement in the demand sector was energy efficiency measures such as the replacement of 70% incandescent bulbs with CFL bulbs. The more aggressive energy policies in the LCA scenario led to the decrees in energy consumption to 2,488 PJ at a growth rate of 2.95%. The household sector saw its share reduced, while the transport sector share of energy consumption increased due to the introduction of LPG for the transport sector (Fig. 3).

The GO scenario, which has a higher level of renewable energy application such as solar PV and efficient use of firewood for the household, had a reduced total energy consumption of 2,249 PJ at a

³ The results of each scenario analysed are presented in Appendix 2.

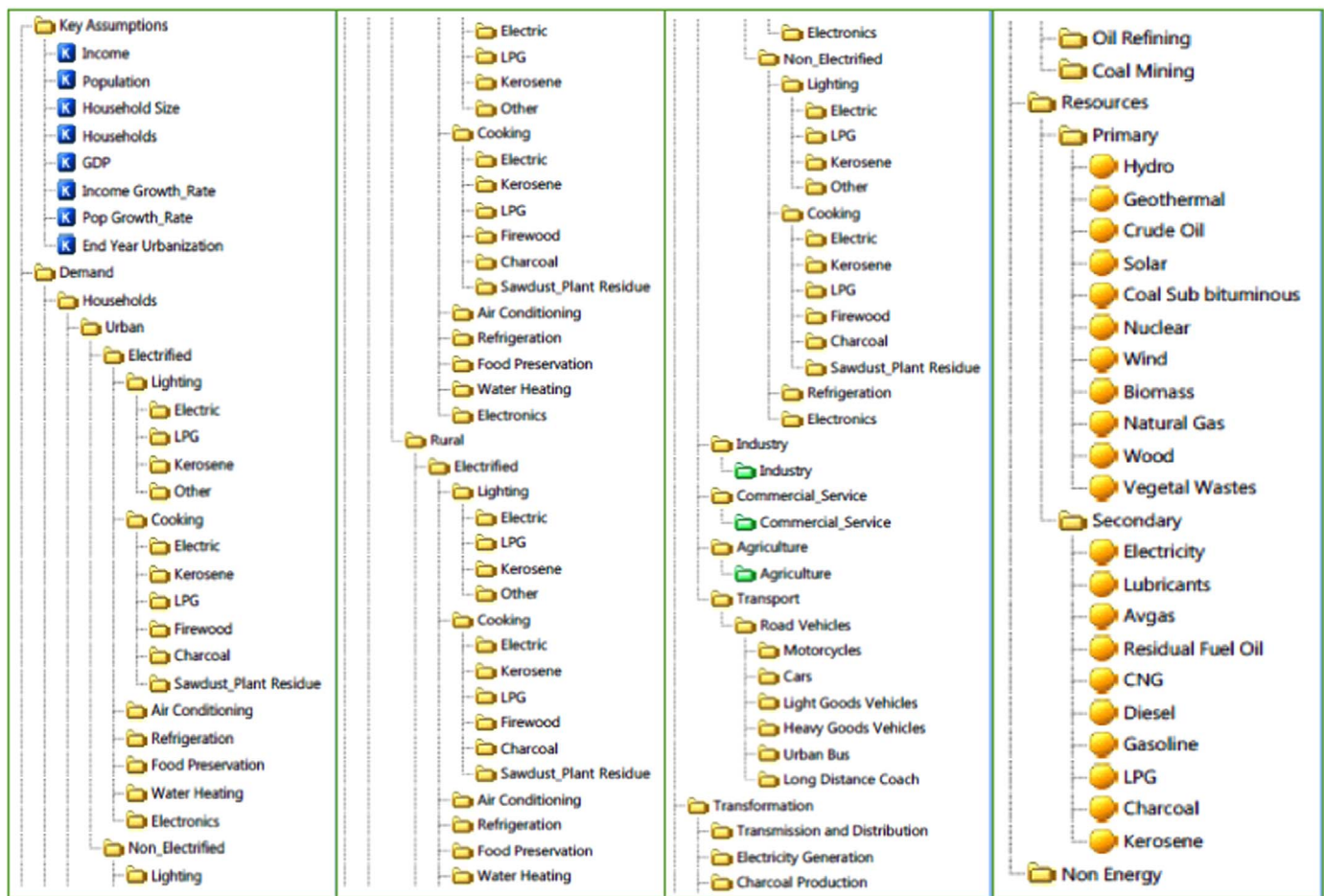


Fig. 1. : The Nigerian LEAP Model branches.

growth rate of 2.61%. As observed in Fig. 3, the household had a slight reduction in energy consumption compared to the LCA scenario, which is due to the phase-out of incandescent bulbs, efficient bulbs, and the provision of solar PV to reduce energy demand.

3.1.2. Final energy use by fuel

The structural changes in the final energy usage in 2020, 2030, and 2040 are shown in Fig. 4 for the four policy scenarios. Under the REF scenario, the energy usage structure will remain almost the same as that in 2010, with a slight increase in electricity consumption to 19%. Biomass, kerosene and gasoline will still dominate the Nigerian energy consumption system by 2040 under the REF scenario. In contrast, under the alternative scenario, the proportion of kerosene consumption will decrease to 23% in the LCM, 21% in the LCA, and 20% in the GO scenario. The decrease of kerosene use in the alternative scenario is seen as a result of energy efficiency improvements in cooking technology (efficient kerosene cooking stoves) and provision of alternative fuel options..

The share of gasoline decreased in the alternative scenarios from LCM (11%) to LCA (10%), while the GO had 9% gasoline use due to the increased popularity of CNG (4%) and LPG (10%) as alternative fuel options for the transport sector. Moreover, the share of clean energy usage, including electricity, LPG, CNG (in the GO), and natural gas will increase rapidly across the alternative scenarios, contributing more than 50% by 2040. Overall, under the LCA and GO scenarios, the energy usage structure will gradually become environmentally friendly in Nigeria, with rapid growth in alternative clean energy. This will play an important role in the realisation of a low-carbon development in Nigeria and GHG mitigation.

3.1.3. Electricity production

To meet the growing demand for electricity, the REF scenario assumes an expansion in electricity generation capacity to be able to generate 179,000 MWh by 2040, as shown in Fig. 5. The electricity mix under the Nigerian government REF scenario by 2040 will include 36% of power generation coming from single cycle gas turbine (SCGT) and 18% from combine cycle gas turbine (CCGT) for on-grid electricity supply, while 18% will be generated from off-grid diesel generators. The contribution of renewable energy technologies is low when compared to the alternative scenarios, especially the LCA and GO scenarios..

The LCA scenario had an electricity supply of 184,000 MWh from the combination of power plants, which comprises of 60% fossil fuel and 40% renewables by 2040. The reduction in the number of SCGT power plants (7%) and the increase in CCGT share will contribute to improved efficiency of gas power plants, since CCGT is more efficient than SCGT. Other fossil fuel power plants with lower emissions include coal integrated gasification combine cycle (IGCC) and circulating fluidised bed (CFB), with shares of 8% and 9% respectively (Fig. 6).

Meanwhile, renewable energy technologies will include large hydro-power (9%), biomass (5%), on-shore wind (2%), and solar thermal and PV systems at 2% each for on-grid power supply as shown in Fig. 6. The mix is further improved with an increased share of renewable power generation of about 70% in the GO scenario. This was intended to increase electricity generation from clean and sustainable sources, while reducing GHG emission levels by 2040.

3.1.4. Greenhouse gas emission

The GHG emission from energy consumption under the four scenarios from 2010 to 2040 is shown in Fig. 7. Under the REF

Table 2
Summary of the features in each scenario.

	REF	LCM	LCA	GO
Driving philosophy	Follows the Government's most-likely-developmental-pathways for the energy system	Driven by cheaper capital, cost, readily available fuels to improve power supply, and moderately reduce energy demand	Motivated by a cleaner fossil fuel power technologies and an aggressive reduction in energy demand	Based on a low-carbon-green growth economy in view of mitigating global climate change, reduce energy poverty, and ensure energy sustainability.
Scenario characteristic	- Current trend of energy consumption continues in all sectors - No supply side diversification of energy source (BAU case)	- Moderate improvement in energy efficiency in all sectors - CFL bulbs to replace 70% incandescent bulbs - Increase in gas power plants, small share of renewables - Introduction biofuel in transport sector - Moderate reduction in electricity T & D and natural gas loss	- Aggressive improvement in energy efficiency in all sectors - LED bulbs to replace 70% incandescent bulbs - Increased capacity of efficient low carbon fossil fuel power plants, increased share of renewables - Introduction of LPG fuel options in addition to biofuel to complement conventional fuels in the transport sector - Improved reduction in electrical and natural gas losses	- Provision of Energy efficiency and solar PV systems for the demand side - Phase-out incandescent bulbs and replace with CFL 60% & LED 40% - Reduction in the share of fossil fuel plant capacity and the increase share of renewables on the supply side - CNG, LPG and biofuel options with a reduced share of conventional fuels - Reduction in electrical and natural gas losses to the barest minimum
General assumptions	Increased electricity access to 100% by 2030			
Scenario-specific assumptions	Demand side: Business-as-usual case in all sectors Supply side: Installed capacity by 2040 = 155283 MW Estimated share of fuel type: Fossil fuel = 90% Renewables = 10%	Demand side: CFL introduction, moderate energy efficiency in appliances Supply side: Installed capacity by 2040 = 170500 MW Estimated share of fuel type: Fossil fuel = 80% Renewables = 20%	Demand side: LED introduction, aggressive energy efficiency in all sectors Supply side: Installed capacity by 2040 = 180500 MW Estimated share of fuel type: Fossil fuel = 60% Renewables = 40%	Demand side: introduction of both CFL and LED with reduction in incandescent bulbs, advance approach towards energy efficiency in all sectors Supply side: Installed capacity by 2040 = 181000 MW Estimated share of fuel type: Fossil fuel = 30% Renewables = 70%

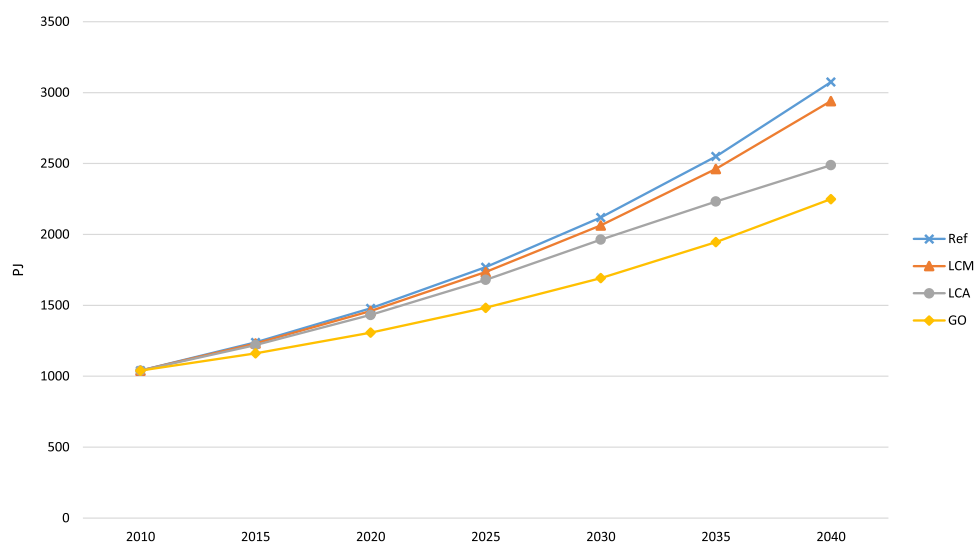


Fig. 2. : Energy consumption under the four scenarios from 2010 to 2040.

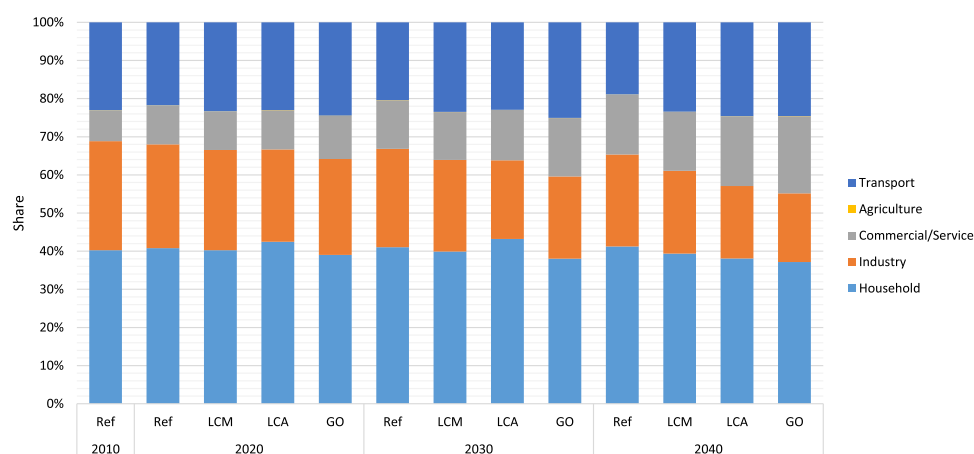


Fig. 3. : Proportion of energy use by sector.

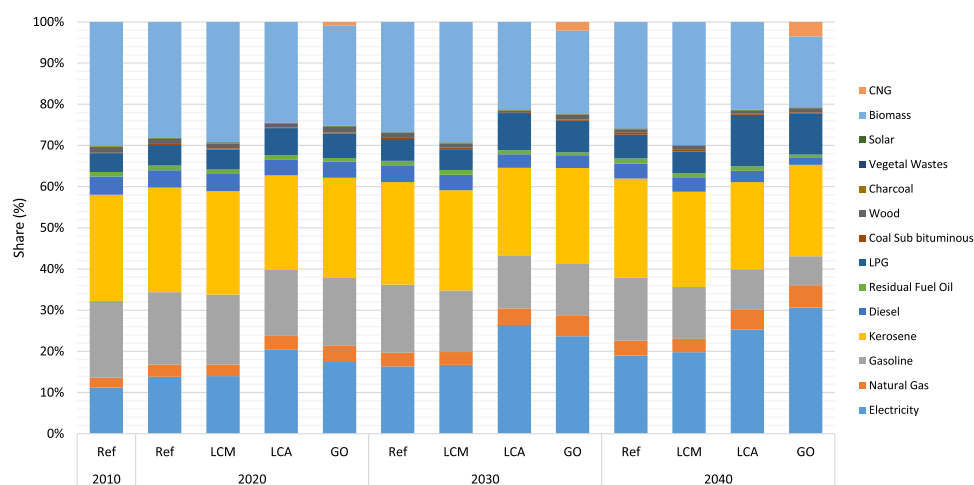


Fig. 4. : Proportion of final energy use by fuel.

scenario, GHG emissions will increase from 50.2 million metric tons of CO₂ equivalent (henceforth Mt Co₂e) in 2010 to 201.2 Mt Co₂e in 2040, with an annual growth rate of 4.74%. The rate of GHG emission can be correlated directly to the level of energy consumption as observed in the REF scenario (compared with Fig. 2), and for the

LCM at 185.4 Mt Co₂e. Under the LCA, emissions will increase to 152 Mt Co₂e in 2040 at a significant growth rate of 3.87%, which is in line with the total energy consumption trend predicted for the LCA in Fig. 3. However, under the GO scenario, the emission will reach a peak value of 124.4 Mt Co₂e in 2040..

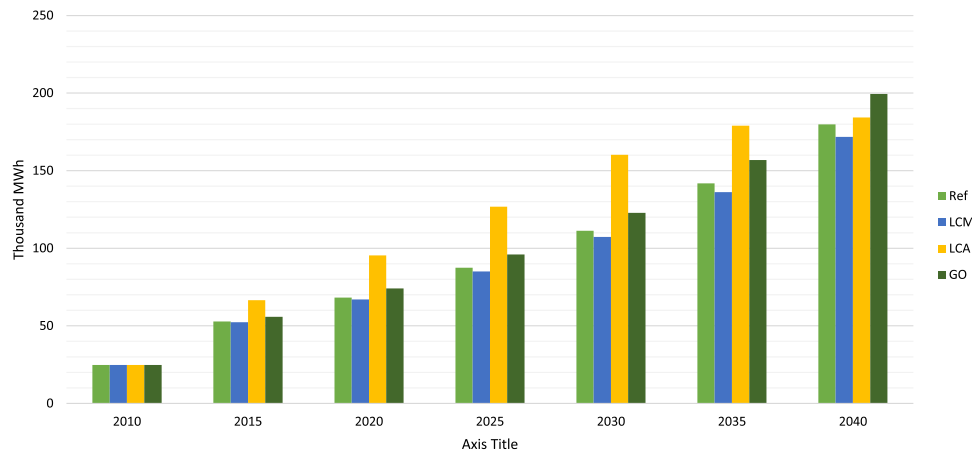


Fig. 5. : Electricity supply projection to 2040 under the four scenarios.

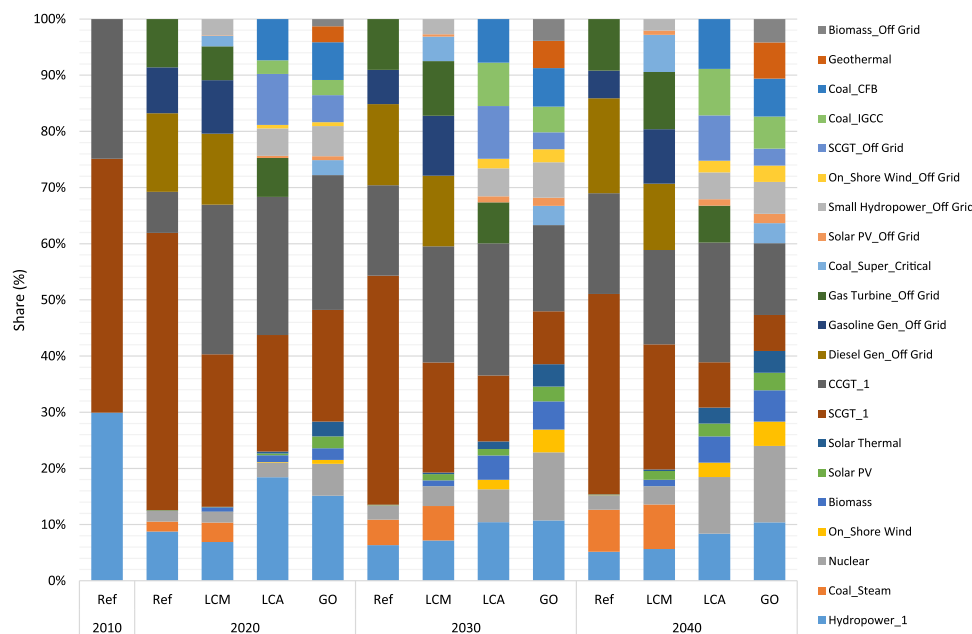


Fig. 6. : Share of electricity generation mix by 2040 under the four scenarios.

The turning point in the GHG emission pathways in the LCA from 2030 may be due to the optimisation of the energy structure. Furthermore, energy intensive sectors such as the iron and steel industries in the GO scenario will experience technology switching,⁴ while electricity usage will be increased in all sectors to reduce fossil fuel consumption.⁵ On the supply side, the increased use of renewable energy in the GO scenario also contributed to the reduction in GHG emissions as can be observed in Fig. 7. This means that to achieve low-carbon development in Nigeria, the introduction and deployment of clean energy technologies should not only focus on demand but also on supply.

3.1.5. Cost-benefit analysis

A cost-benefit analysis for the scenarios were analysed in this study and the results are presented in Table 3, which also highlights Net

⁴ This will be the switching to electric arc from conventional blast furnaces and other efficiency improvement in the industrial sector.

⁵ Kerosene, firewood, charcoal, and biomass consumption are expected to reduce in the demand sector, since the electricity supply will be increased and the use of electricity for cooking and lighting will be encouraged.

Present Value (NPV) of alternative scenarios relative to the reference scenario. NPV represents all discounted costs⁶ and benefits in one scenario minus another.⁷

From Table 3, we can observe that under the demand variable (or demand side), the cost for the implementation of the LCM scenario is USD 36.62 million, while LCA and GO scenario was USD 31.32 and USD 30.61 million respectively. In implementing the policies in the LCM scenario, USD 2.73 million will be saved by the Nigerian government (as benefit), while USD 38.93 million would have been spent in achieving the policies in the transport sector and USD 42 million in the households sector. In the LCA scenario, more would be spent for households' energy efficiency policies (USD 560 thousand), while USD 38.11 million will be spent in the transport sector.

The costs in the transport sector will include the introduction and increase of LPG vehicles, energy efficiency, and fuel switching. The transformation sector showed various levels of costs which varied

⁶ This study assumed the discounted cost to be 5% in 2010.

⁷ This is summed up across all the years of the study; and in this study, it was summed from 2010 to 2040.

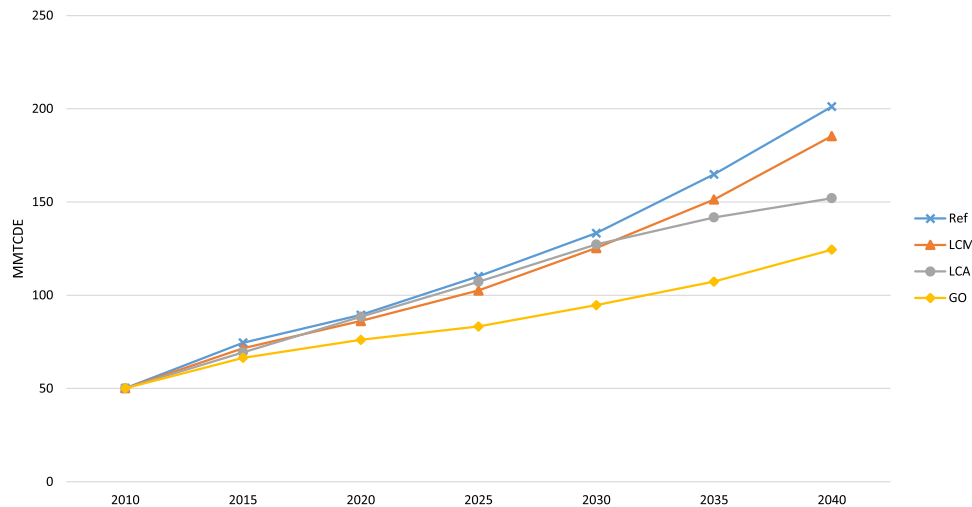


Fig. 7. : GHG emissions under the four scenarios from 2010 to 2040.

Table 3

Results of the cost-benefit analysis.

Cumulative Costs & Benefits: 2010–2040. Relative to Scenario: Reference.			
Discounted at 5.0% to year 2010. Units: Billion 2010 U.S. Dollar			
	Low Carbon Moderate	Low Carbon Advance	Green Optimistic
Demand	36.62	31.32	30.61
Households	0.42	0.56	0.57
Industry	–	–1.61	–2.35
Commercial_Service	–2.73	–5.74	–9.56
Agriculture	–	–	–
Transport	38.93	38.11	41.95
Transformation	15.65	36.12	81.62
Transmission and Distribution	–	–	–
Electricity Generation	15.65	36.12	81.62
Charcoal Production	–	–	–
Oil Refining	–	–	–
Coal Mining	–	–	–
Resources	1,640.19	23,790.23	41,269.17
Production	–209.43	44.14	–303.14
Imports	1,849.62	23,746.09	41,572.31
Exports	–	–	–
Unmet Requirements	–	–	–
Environmental Externalities	–4.02	–24.05	–34.48
Non Energy Sector Costs	–	–	–
Net Present Value	1,688.44	23,833.61	41,346.92
GHG Savings (MMTCDE)	221.73	332.14	952.00
Cost of Avoiding GHGs (U.S. Dollar/TCDE)	7,615.00	71,757.93	43,431.71

*Tonne per Carbon Dioxide Equivalent.

across scenarios. The LCM scenario had a lower cost in terms of implementing the strategies,⁸ while LCA had a higher cost of USD 36.12 million. Environmental externalities⁹ presented a benefit in all

⁸ These strategies include the building and capacity expansion of power plants that are not in the REF scenario. For some power plants in the REF scenario that are in the alternative scenarios, the differences are taken which is the cost of the expanded capacity of the alternative scenario which is the differences between the two scenarios. Please note that the strategies also include the cost of improvement in T & D.

⁹ Environmental externalities refer to the economic concept of uncompensated environmental effects of production and consumption that affect consumer utility and enterprise cost outside the market mechanism. As a consequence of negative externalities, private costs of production tend to be lower than its “social” cost. It is the aim of the “polluter/user-pays” principle to prompt households and enterprises to internalise externalities in their plans and budgets [46].

the alternative scenarios, with the GO scenario having the most benefit of USD 34.48 million, followed by the LCA scenario with a benefit of USD 24.05 million. The overall NPV for the alternative scenarios are USD 1.69 billion for the LCM, USD 23.8 billion for the LCA, and USD 41.4 billion for the GO scenario. This overall NPV is the cost of how much more the alternative scenario¹⁰ costs versus the REF scenario. Lastly, the cost of avoided¹¹ GHG emissions was higher in the LCA (USD 71.8/TCDE) scenario, than in the GO (USD 43.4/TCDE).

3.1.6. Future energy system

Notably, with the accelerated population, urbanization and improvement of living standards in Nigeria, the energy consumption in the household, industry, commercial/service, and transport sector is predicted to increase greatly in the future. This will directly affect the total energy consumption and energy system in the future from 2010 as shown in Fig. 8, to the four scenarios as shown in Fig. 9 (REF), Fig. 10 (LCM), Fig. 11 (LCA), and Fig. 12 (GO).....

The observed changes showed that as various alternative fuels becomes popular and acceptable in Nigeria, the refinery capacity in Nigeria which currently stands at a total of 445,000 barrels per day (bbl/d) will not be sufficient enough to meet the growing energy demand. This implies that natural gas products such as LPG of 114.9 PJ will have to be imported to support the low refinery capacity in Nigeria for the GO scenario, while CNG will be refined locally to meet the transport demand of 28.4 PJ.

Crude oil product import such as gasoline (453 PJ) is expected to decrease across the alternative scenarios as more alternative fuel options such as biofuel (in LCM), LPG (in LCA), and CNG (in GO) becomes commercially available. The same goes for kerosene import in the REF (632.1 PJ), which decreased across scenario, and had the lowest import in the GO (391 PJ). The increased use of improved stoves for charcoal, firewood and other biomass produce will not only complement kerosene use, but also ensure the efficient utilization of Nigeria's energy resources with a reduced GHG emissions. On the supply side, the import of nuclear fuel will increase from the REF scenario (46.5 PJ), to the LCA (186 PJ) and GO (262.1 PJ) by 2040, which is correlated with the power plants shown in Fig. 6. The energy system in general under the alternative scenarios show great potentials for the reduction of GHG emissions under various level of policy implementations in each scenario. The “bottle-neck” will be the availability of funds for the implementations of the proposed policies

¹⁰ Please note that it is one alternative scenario versus the REF scenario (i.e. LCM vs. REF, LCA vs. REF, etc.).

¹¹ This is given by dividing the NPV by the TCDE avoided.

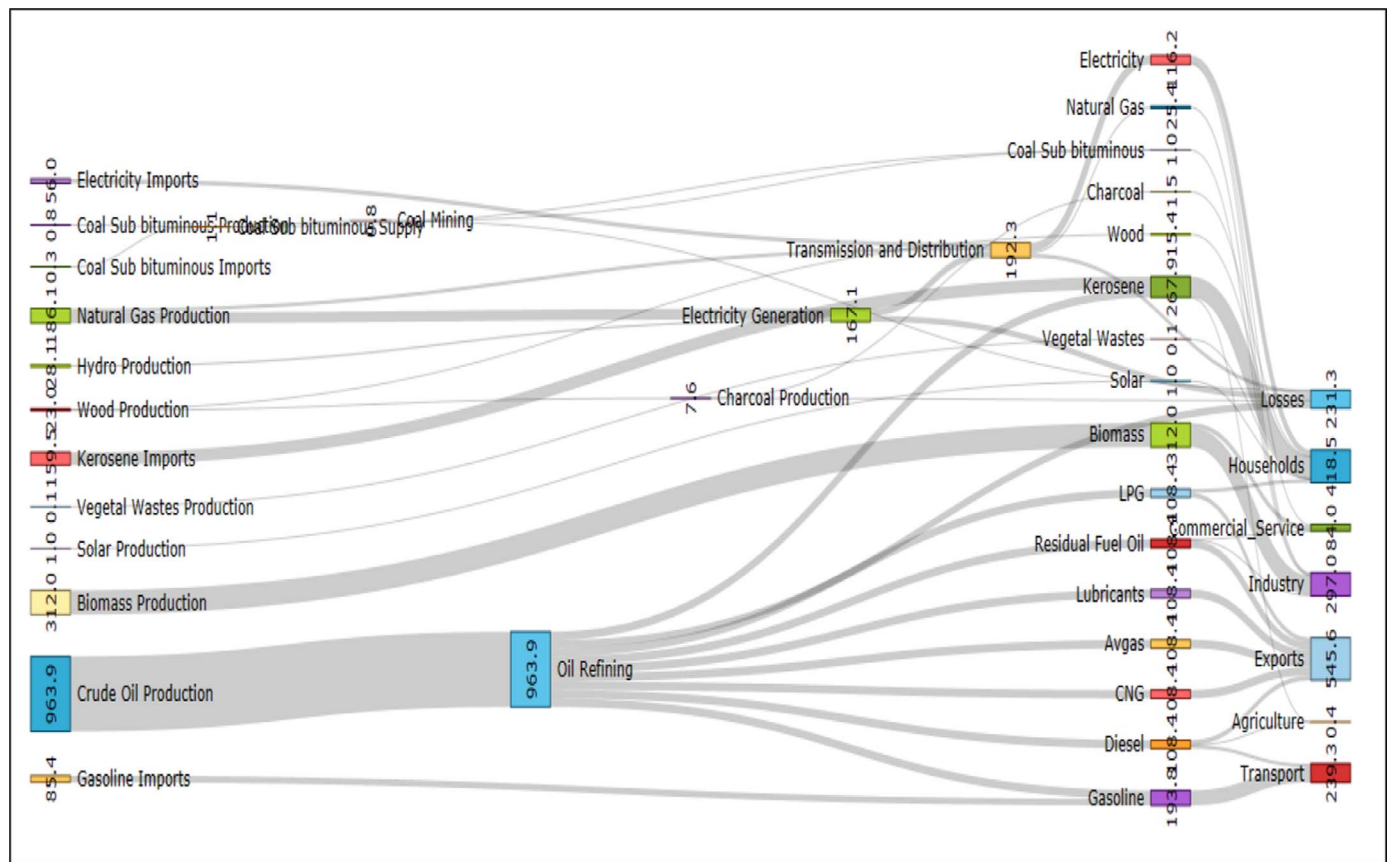


Fig. 8. : Nigeria energy system in 2010.

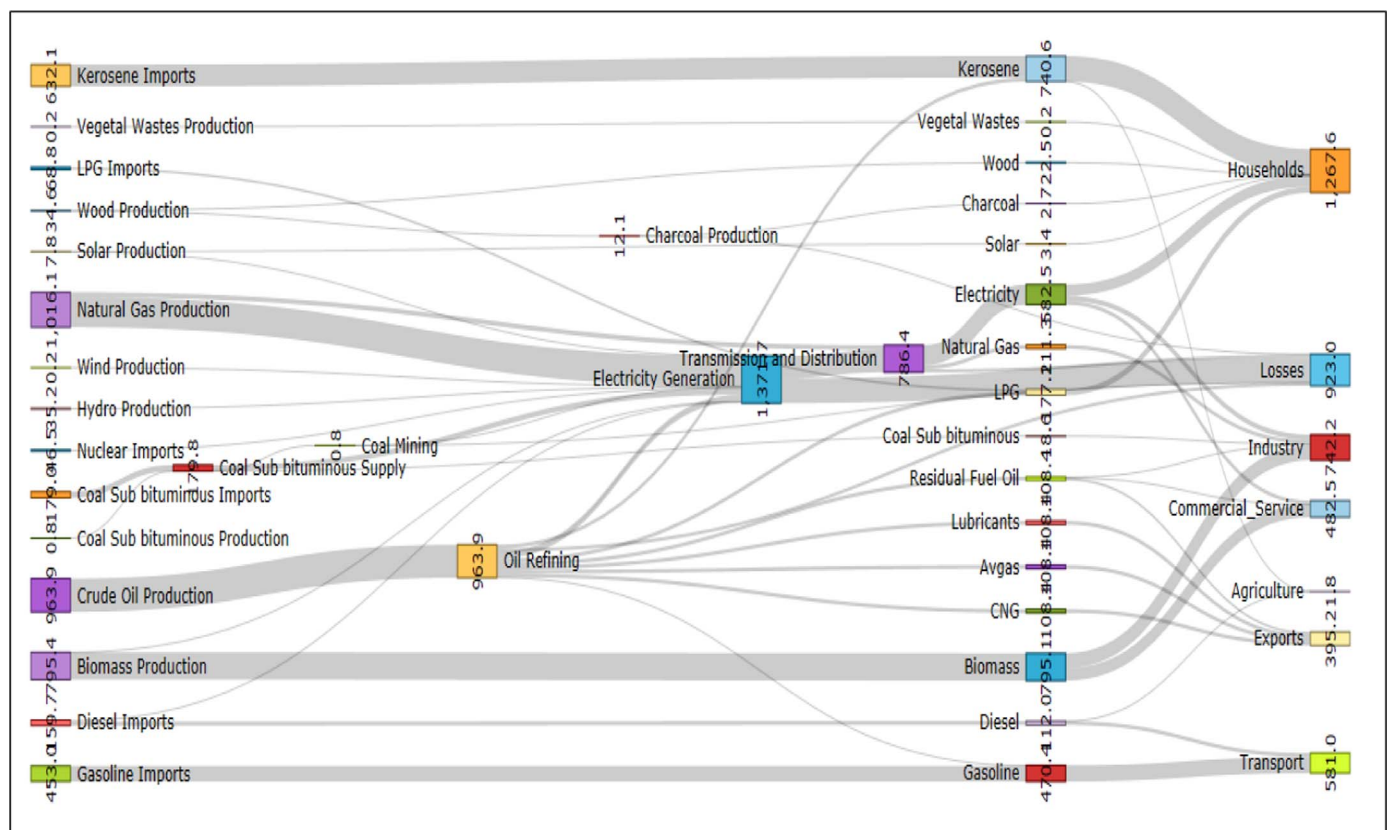


Fig. 9. : Nigeria energy system in 2040 under the REF scenario.

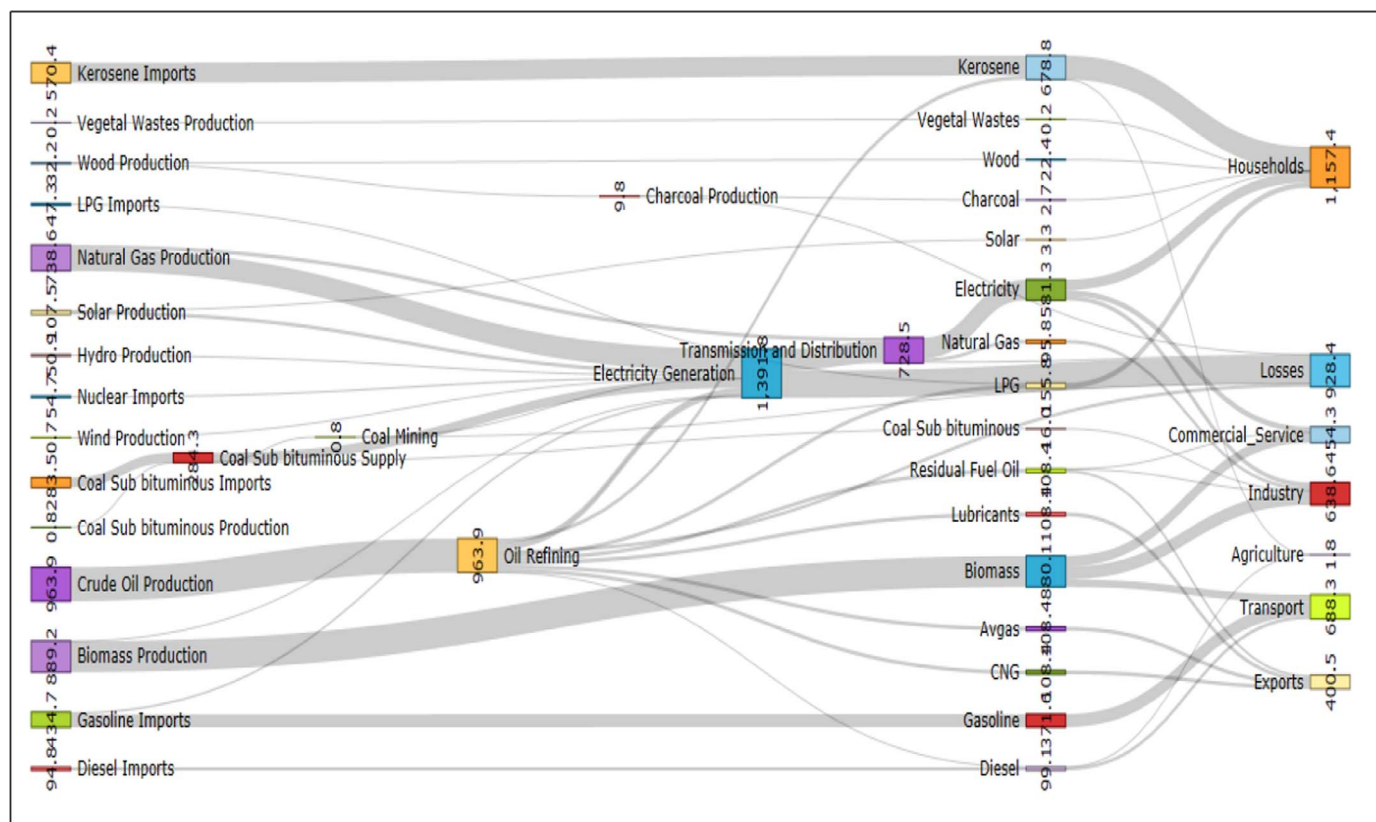


Fig. 10. : Nigeria energy system in 2040 under the LCM scenario.

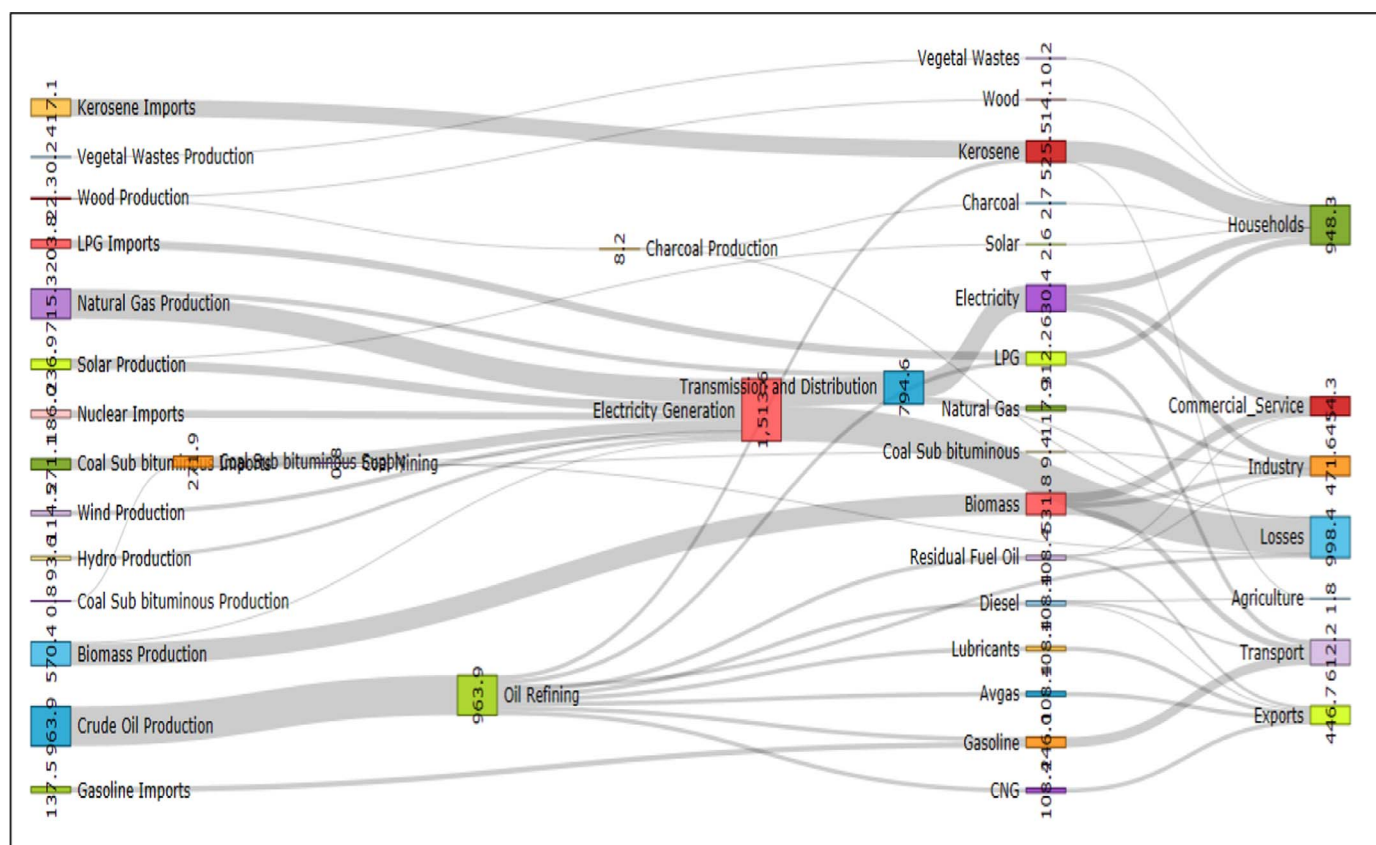


Fig. 11. : Nigeria energy system in 2040 under the LCA scenario.

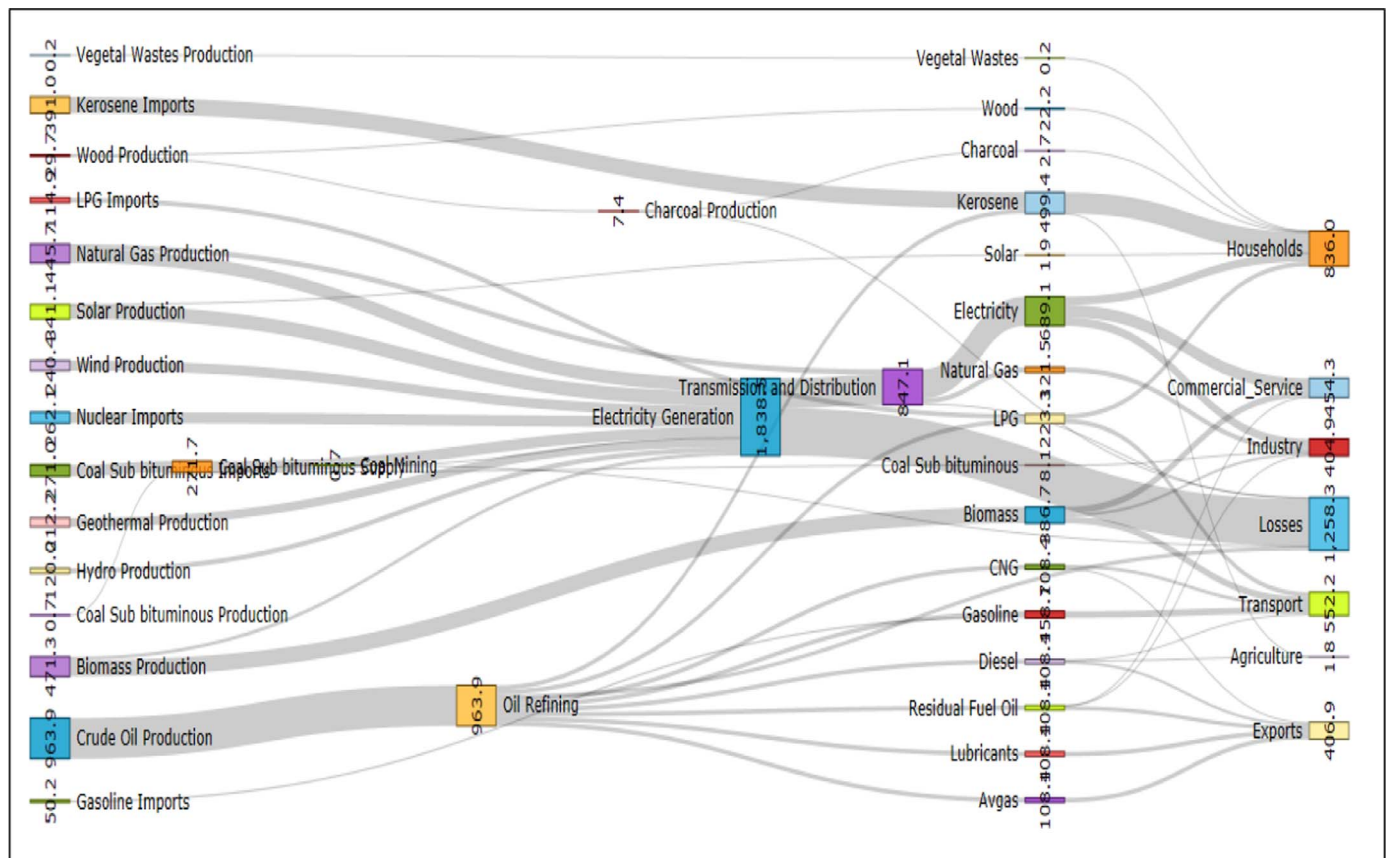


Fig. 12. : Nigeria energy system in 2040 under the GO scenario.

and strategies, and the solution to this will be discussed in subsequent sections.

3.2. Discussion

Nigeria as a signatory to the UNFCCC and Kyoto Protocol will have to make international commitments in promoting low-carbon development by meeting its reporting obligation to the UNFCCC and reducing greenhouse gases (GHG) emissions within the context of poverty reduction and economic growth [14]. Among the four policy scenarios, the alternative scenarios will achieve some improvement in energy and GHG reduction. Under the LCA and GO scenarios, the emission will be 7% and 11% lower than the REF scenario by the end of 2040. Obviously, the GO scenario shows a more favourable development trajectory towards low-carbon development for Nigeria, demonstrating that more carbon emission reductions are possible by implementing additional policies and strategies. Thus, specific measures can be recommended for certain sectors of Nigeria's economy to achieve and sustain low-carbon development.

The residential sector should place emphasis on improving the efficiency of energy technologies such as kerosene, charcoal, and firewood cooking stoves. The ECN stressed the need for the massive deployment of these efficient cooking stoves for the Nigerian households in its national energy policies, but efforts to popularise the energy technologies have been slow. To complement this, LPG and solar cookers should be introduced into the household energy mix. Agbo and Oparaku [1] highlighted some pressing policy issues preventing the deployment of these energy technologies and alternative fuels including poor research and development (R & D), pilot and demonstration projects, institutional framework, investment promotion, incentives, and protections.

However, if the Nigerian government intends to achieve a reduction

in energy demand and GHG emissions as observed in the LCA and GO scenarios, these challenges need to be addressed. On improving the efficiency of household electrical appliances, the emphasis should be on implementing energy standards for refrigerators, air conditioners, and electric stoves; and more importantly, phasing out incandescent bulbs and replacing them with LED and CFL bulbs.

Some measures, such as “Phase out Incandescent Light Bulbs” (POILB), should be employed.¹² Many countries around the world have employed this policy measure including China, which started the ban in 2012 but extended it to 2016¹³; India, which also started the ban on incandescent bulbs in 2012¹⁴; Israel has been phasing out incandescent bulbs since 2012¹⁵; the United Kingdom in 2011¹⁶; and all European Union (EU) countries have until 2016 to completely phase out incandescent bulbs before 2016.¹⁷ Other countries include Canada, who made a move in 2007,¹⁸ while most states in the United States (US) have completely phased out incandescent bulbs since 2007 and others will have in 2018.¹⁹ Some countries implemented new energy standards and phased out incandescent bulbs like Argentina in 2012²⁰

¹² See www.energyrating.gov.au/products-themes/lighting/lighting-and-phase-out-general-information/incandescent-light-bulbs-phase-out/

¹³ www.reuters.com/article/2011/11/05/us-china-light-bulbs-idUSTRE7A40MV20111105

¹⁴ www.treehugger.com/interior-design/india-to-phase-out-400-million-incandescent-lightbulbs-by-2012-replace-with-cfls.html

¹⁵ www.energy.gov.il/LightBulb/Pages/GxmsMniMiniSiteLightBulb.htm

¹⁶ www.news.bbc.co.uk/2/hi/uk_news/7016020.stm

¹⁷ www.greenpeace.org.uk/blog/climate/eu-ban-inefficient-light-bulbs-eventually-sort-20081212

¹⁸ www.reuters.com/article/2007/04/25/us-lightbulbs-env-idUSN2529253520070425

¹⁹ www.leginfo.ca.gov/pub/07-08/bill/asm/ab_1101-1150/ab_1109_bill_20070223_introduced.html

²⁰ <http://dx.doi.org/10.1016/j.rser.2016.09.118>

and Mexico, Malaysia, and South Korea²¹ in 2014. The Nigerian government can also stimulate the residential sector to improve household efficiency through the provision of subsidies for consumers who purchase and use efficient appliances, as well as renewable energy sources such as solar thermal and PV systems.

For the commercial/service sector, similar policy scenario for the household sector holds for electrical appliances and lighting bulbs. Furthermore, the provision of adequate electricity supply will ensure the proper shift towards the use of electricity as final energy consumption, instead of fossil fuels such as diesel, gasoline, biomass, etc. The same stimulus policy for efficiency improvement and renewable energy integration in the demand side will aid in the transition toward low-carbon development. Public services, such as hospitals, will now have a means of building a sustainable and unlimited power supply to carry out their social duties, while reducing the need for dependence on diesel and gasoline for private power generation.

For the industrial sector, optimisation of the production structure and improvement in energy efficiency are vital in achieving low-carbon development. This can be promoted by energy auditing in industries, as in the study carried out by Olayinka and Oladele [33]. In their study, they found out that consumption by electric motors accounts for 40–47% of the total electricity consumed in most industries in Nigeria. Others, such as boilers and heaters, account for 65% of the total energy consumed. The inefficiencies in energy use were attributed to poor housekeeping of air conditioners, refrigeration equipment, wheel electric motors, and lack of switching off electric bulbs during the day. These inefficiencies, if addressed, will result in the reduction of about 30% of energy demands in the industrial sector.

In addition, electric arc blast furnaces for iron and steel industries should replace the conventional ones by 2040. The introduction of these furnaces has great potential for the reduction of energy/electricity consumption and GHG emissions due to their high level of energy efficiency. This technology switching will increase the share of electricity in the industrial final energy mix, while reducing fossil fuel consumption. The implementation of this policy on a short timescale may be difficult, but can be tackled through the provision of financial incentives, subsidies, and tapping into global energy-saving funds such as clean development mechanism (CDM) and global climate funds (GCF) to help promote the industrial transformation.

For the transport sector, the use of public transport systems such as the bus rapid transport (BRT) system already in Lagos, Nigeria and light rail in some parts of Nigeria should be encouraged and popularised by the Nigerian government. The implementation of alternative vehicle fuels such as CNG, LPG, and biomass will reduce the dependence on high GHG-emitting gasoline and diesel fuels. The South Korean BRT system is comprised of mainly CNG-powered buses. This move was to ensure a reduction in GHG emissions. In the Korean transport sector, LPG is another alternative fuel besides CNG that is intended to provide consumers with low GHG-emitting fuel options. The LCA and GO scenarios proved that this could be replicated in the Nigerian transport sector if the government is ready to implement some policies such as the Autogas Incentive Policies²² (AIP) to stimulate commuters. The AIP has contributed to the increased use of LPG for transportation in most countries around the world.²³

In terms of the energy supply sector, the reduction in T & D losses should be promoted for the electricity sector as well as natural gas supply, which is used to power SCGT and CCGT power plants. However, the Nigerian government should encourage construction of

CCGT power plants over SCGT, which is less efficient. For coal power plants, more efficient combustion technologies such as IGCC, CFB, and supercritical should be given more preference over the conventional coal steam power plants. Although coal power plants are higher GHG-emitting energy technology, great potential exists for carbon capture and storage (CCS) technology in future. The Nigerian government should focus more on development of nuclear and renewable energy sources, which are low-carbon technologies that are vital for the attainment of a low-carbon development before 2040.

The government can increase the capacity of nuclear power plants to contribute more to the electricity mix as observed in the LCA and GO scenarios, instead of the small share in the REF scenario. This should be matched with renewable sources such as solar, wind, hydro, biomass, and geothermal.²⁴ This actually disproves the Ibitoye [23] study, where his bottom-up approach using the LEAP model suggested that the most likely development pathway for Nigeria to generate electricity to meet the millennium development goals (MDGs) was through natural gas alone. This study, therefore, provides electricity generation alternatives that will not only meet the increasing energy demand, but also with lower GHG emissions due to using sustainable energy resources.

Although this is the first study that applied a bottom-up approach to project the pathway for the complete energy system in Nigeria, great care should be taken to understand the term “low-carbon development” in the Nigerian context. According to the Conference of Parties to the UNFCCC, the concept of low-carbon development strategies (LCDS) is a common but differentiated approach to meet the overall emissions reduction objectives: “All countries shall prepare Low Emission Development Strategies...nationally-driven and represent[ing] the aims and objectives of individual Parties in accordance with national circumstances and capacities” (Cancun Agreement).²⁵ This means that for a developing country like Nigeria, the path towards low-carbon development should be based on the nation's circumstances and ability to go the extra mile in attaining a low-carbon economy. However, “low-carbon” is a futuristic development that must be pursued, not necessarily as the goal, but rather to test our efforts towards low-carbon development. The low-carbon developmental pathway in Nigeria is not static, but dynamic; therefore it needs to be constantly amended at different stages and periods in time. This study provides some likely developmental pathways through the developed scenarios for the Nigerian energy system, and explored the impact of future energy policies and strategies that will be required on the path to a low-carbon transition.

4. Conclusions and policy implications

The prospects of a low-carbon development in Nigeria's future energy system will depend on the energy policies initiated and implemented in the future. In this study we developed an energy model for Nigeria which considered vital factors capable of influencing energy policies in future. The four scenarios developed in the model include a reference scenario and three alternative scenarios. The alternative (LCM, LCA, GO) scenarios were influenced with various levels of policies and strategies to check different low-carbon development pathways for Nigeria from 2010 to 2040 using the LEAP modelling tool. The results showed that the strategic policies implemented in the Nigerian LEAP model will have a significant impact on energy consumption and GHG emissions.

Under the REF scenario, energy demand in 2040 will be 3,075 PJ with a growth rate of 3.68% with a corresponding increase in GHG

²¹ www.yonhapnews.co.kr/economy/2013/07/16/0302000000AKR20130716057151003.H-TML

²² The AIP is promoted by the government through lowering fuel tax vis-à-vis gasoline and diesel, lowering vehicle taxes or conversion subsidies, employing traffic measures, and removing barriers (parking restrictions, etc.).

²³ www.bpnews.com/index.php/publications/magazine/current-issue/305-world-autogas-demand-jumps-57-in-past-decade-incentives-play-key-role

²⁴ Although geothermal energy has not been widely popularised in Nigeria, studies [2,21,31,34] and [32]) have analysed various parts of Nigeria for geothermal resources, while Kurowaska and Krzysztof [25] highlighted that the geothermal resources in Nigeria were enough to be exploited for power generation.

²⁵ See <http://cancun.unfccc.int/>

emissions of 201.2 Mt Co₂e. Policy scenarios such as the LCM, LCA and GO scenarios had lower energy demands of 3.53%, 2.95% and 2.61% respectively. Similarly, GHG emissions were generally reduced across the policy scenarios. To achieve such large reductions in energy demand and GHG emissions, strategic policies and technological switching is not enough, although they contributed to the reduction in energy consumption and GHG emissions in Nigeria. The Nigerian government will need to do more to promote low-carbon energy technologies and energy efficiency measures through policy interventions and incentives.

Since the focus of this study was to transit Nigeria towards a low-carbon economy, the optimisation of the energy structure was focused on shifting from fossil fuel to clean energy, as in the LCA and GO, which will play a significant role in GHG reduction in Nigeria. The cost-benefit analysis results showed that the implementation of the policies in the LCM will amount to a cost of USD 1.69 billion, the LCA policies will cost USD 23.8 billion, while the GO policies will cost USD 41.1 billion. Although the costs are high due to investments in energy efficiency, fuel switching, etc., these costs are somewhat offset by the energy savings on the demand side, production, and environment externalities. Overall, the household and transport sector are key areas for low-carbon development in Nigeria, while the industrial and commercial/service sectors also hold great potential in the reduction of energy consumption and GHG emissions in the coming years.

In conclusion, for Nigeria to achieve low-carbon development, it is import for the country to explore various policy options relevant to the reduction of energy demand and GHG emissions with clean energy resources. However, it is not expected to be easy for Nigeria to achieve low-carbon development, as some challenges cannot be ignored. For example, the deployment of efficient cooking stoves and other clean energy technologies in Nigeria has proved challenging for the government, due to the high cost of the technologies for the consumers on the demand side. On the supply side, the cost of renewable energy is still very high as it requires huge capital investment, and this increases the preference for fossil fuels such as natural gas. This could be addressed if the Nigerian government implements some fiscal incentives such as tax relief and exception for clean energy technology imports, subsidies and loans for the purchase of renewable energy, and public awareness to raise interest on the benefit of clean energy technologies.

Policy implications of the study include: (i) It is important for Nigeria to improve on energy efficiency in both the supply and demand sides through the application of modern energy technologies and practice, as this will have an impact on Nigeria's future energy conservation and GHG reduction. (ii) The government should make provisions for the development and deployment of alternative fuels in the demand and supply sides, as this will reduce dependence on a particular energy source such as biomass or crude oil. (iii) The government should make incentives such as investment tax credits, low cost loans for power generation, and increased feed-in-tariff options. (iv) The establishment of energy and environmental policies should support clean energy, as this will accelerate the transition to a low-carbon economy in Nigeria. (e) Low-carbon development should be seen as a step in the right direction, and this should be done according to the ability, capacity, and capability of Nigeria. This research is not without limitation, as some transport variables such as railway, airway, and seaway transport system were excluded due to lack of available data. Additionally, the alternative scenarios do not exhaust the possibility of providing solutions to a low-carbon development in Nigeria. Thus, more policy scenarios can be developed in future studies which will include the complete transport variables. Another improvement will be to consider the impact of climate change on energy demand and supply in the developed model, which will eventually update the present study with further policy implications.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.rser.2016.09.118>.

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